Subgrid Recipe for Multiphase Circumgalactic Medium Simulations

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Abstract

The circumgalactic medium (CGM) contains a significant fraction of a galaxy's baryons and plays a crucial role in galaxy evolution. The cold (~10^4 K) component of the CGM exists as dense, small-scale cloudlets embedded in a hotter, diffuse phase. However, simulating the interactions between these phases remains a significant challenge due to the extreme disparity in spatial scales of these cold gas structures and their impact scales on the associated system's evolution. While high-resolution studies have advanced our understanding of cold gas dynamics, fully resolving multiphase CGM in cosmological simulations is computationally infeasible. To address this, we develop a subgrid model (STW: **S**imple **M**odel that **W**orks), a novel multi-fluid framework designed to address these resolution limitations. Our model enables realistic CGM evolution within practical computational limits, with scope of improving galaxy formation simulations. We use PLUTO simulation package for the implementation and testing of STW model.

Traditional Hydrodynamical Simulations

Traditionally , in hydrodynamic simulations, the evolution of the gas fluid is described by the conventional hydrodynamic conservation equations for mass, momentum, and energy:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}^{\mathbf{T}} + \mathbf{I}p) = 0$$

$$\frac{\partial}{\partial t}\left(\frac{1}{2}\rho \mathbf{u}^{\mathbf{2}} + \frac{p}{\gamma - 1}\right) + \nabla \cdot \left[\left(\frac{1}{2}\rho \mathbf{u}^{\mathbf{2}} + \frac{p\gamma}{\gamma - 1}\right)\mathbf{u}\right] = (q^{+} - q^{-})^{cool}_{net}$$

ρ: physical density of fluid, u : fluid velocity, p: pressure in a simulation cell.

q^{-} : change in energy density per unit time due to cooling , q^{+} : redistributed energy density that heats up the cell to ensure global equilibrium

$$q^{-} = n^2 \Lambda(T) \qquad q^{+} = \rho \frac{< q^{-} >}{< \rho >}$$

STW: Subgrid Model

We follow a semi-explicit two fluid approach where hot phase and a misty cold phase are spatially coexisting and co-moving in a simulation cell. The model evolves cell average properties, and a tracer equation keeps track of cold mass fraction in a cell.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}^{\mathbf{T}} + \mathbf{I}p) = 0$$

$$\frac{\partial}{\partial t}\left(\frac{1}{2}\rho \mathbf{u}^{\mathbf{2}} + \frac{p}{\gamma - 1}\right) + \nabla \cdot \left[\left(\frac{1}{2}\rho \mathbf{u}^{\mathbf{2}} + \frac{p\gamma}{\gamma - 1}\right)\mathbf{u}\right] = (q^{+} - q^{-})^{cool}_{net}$$

Here, ρ is the volume-weighted average density of the two phases.

$$\frac{\partial (f_m^{cl}\rho)}{\partial t} + \nabla \cdot (f_m^{cl}\rho \mathbf{u}) = S_{rad} + S_{mix}$$

f_m^{cl}: cold mass fraction in a cell

$$S_{rad} = \frac{q^{-}}{\left(\frac{3k_b(T_h - T_{cl})}{2\mu m_p}\right)} \qquad q^{-} = \frac{f_v^h \rho_h^2 \Lambda(T_h)}{(\mu m_p)^2}$$

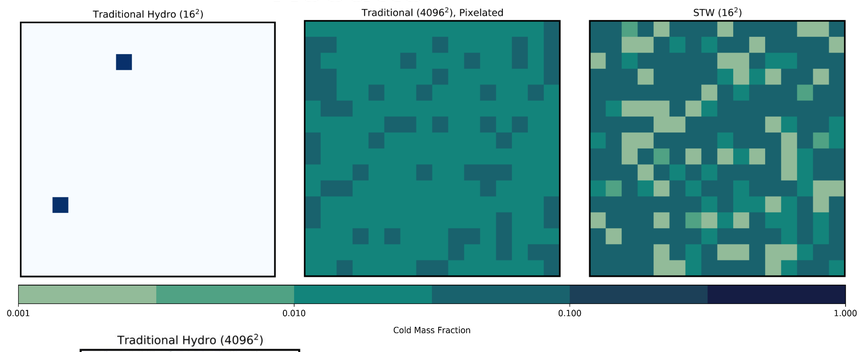
$$\rho_{cl} = \frac{p\mu m_p}{T_{cl} k_b}, \quad \rho_h = \frac{\rho - f_v^{cl}\rho_{cl}}{(1 - f_v^{cl})}, \quad T_h = \frac{\rho_{cl} T_{cl}}{\rho_h}$$

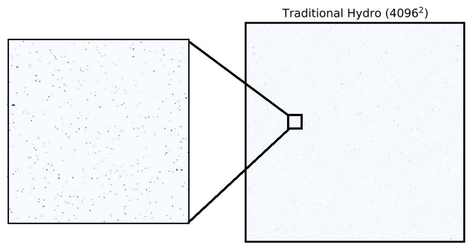
$$S_{mix} = \rho \left[\frac{\left[(f_m^{cl})(1 - f_m^{cl})^2\right]^{3/2}}{\chi \sqrt{t_{destroy} t_{cool,mix}}} - \frac{\left[(f_m^{cl})^2(1 - f_m^{cl})\right]^{3/2}}{\chi^{1/2} t_{destroy}}\right] \qquad t_{destroy} = \frac{1}{\left|\frac{\Delta u + (\Delta u)^T}{2} - \frac{1}{3}(\Delta u) I\right|},$$

$$\chi = \frac{\rho_{cl}}{\rho_h}; \quad T_{mix} = \sqrt{T_h T_{cl}}, \quad \rho_{mix} = \sqrt{\rho_h \rho_{cl}} \qquad t_{cool,mix} = \frac{(\mu m_p)^2}{(\gamma - 1)\Lambda_0} p \frac{T_{mix}^{2/3}}{\rho_{mix}^2}$$

Results

Thermal Instability Tests

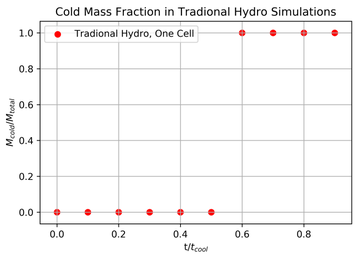


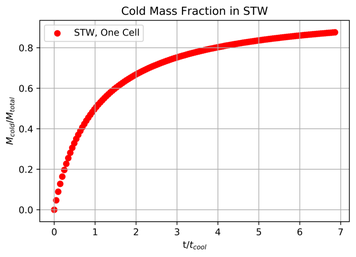


Above: The distribution of cold gas formed through thermal instability is illustrated in a set of idealized 2D simulations (Traditional vs STW) of a 16 × 16 kpc region of CGM.

Below: The evolution of cold mass fraction in s single simulation cell with radiative cooling in traditional and STW model.

Single Cell Simulation

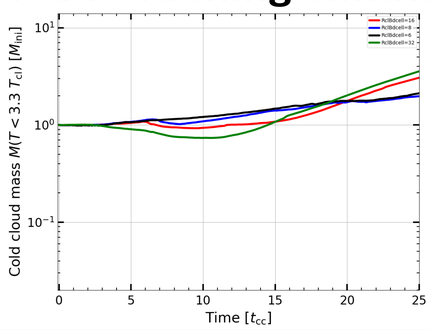


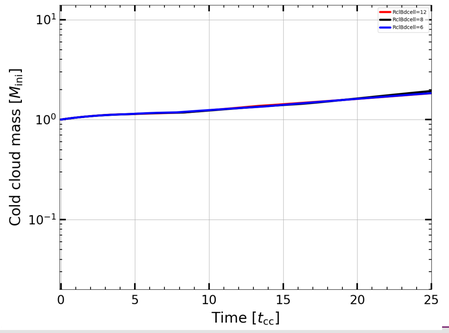


Cloud Crushing Simulations

A cold, dense cloud is embedded in a hot gas flow, leading to a combination of hydrodynamical instabilities such as the Kelvin-Helmholtz (KH) and Rayleigh-Taylor (RT) instabilities.

The left plot shows cold mass evolution in a traditional simulation of different resolutions while right one shows the same for STW model.





Conclusion

Our findings highlight the importance of physically motivated subgrid models in bridging the resolution gap between small-scale CGM physics and large-scale cosmological simulations. By capturing unresolved multiphase interactions, our semi-multifluid approach provides a more realistic and computationally efficient representation of cold CGM gas dynamics.

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